

LAGUNA UNIVERSITY
College of Computing Studies
Laguna Sports Complex, Brgy. Bubukal, Sta. Cruz, Laguna

SYSTEM INTEGRATION AND ARCHITECTURE 2
& MOBILE APPLICATION DEVELOPMENT
FINAL PROJECT - PROPOSALS

GROUP NO.: 2

BLOCK/SPECIALIZATION: BSIT BA -3B

STUDENT MEMBERS (LN, FN, MI)

Role	Student name	Primary responsibility
Member 1	MONAHAN, Lance Bradly A.	Backend
Member 2	VENTURA, Jane Angelie M.	Frontend

SIA 2 SUBJECT TEACHER: Kurt Robin San Jose

MOBILE APP DEVELOPMENT TEACHER: Deiter Ronghelo Palma

DATE SUBMITTED: _____

This packet presents three client-centered project alternatives. Each proposal must demonstrate a coherent Web + Mobile + Backend/Database + ESP32 integration when the ESP32 provides genuine sensing or control value. A large feature list is not a substitute for a clear workflow, shared data model, feasible architecture, and client validation.



PROPOSAL PREPARATION GUIDE

Complete all required fields. Use concise, evidence-based writing. Attach client interview notes, process photos only when permitted, preliminary diagrams, and supporting references in the appendices.

Minimum proposal requirements

- Three distinct titles addressing a verified client problem.
- A real prospective client and an authorized contact person.
- One end-to-end workflow shared by Web, Mobile, Backend/Database, and ESP32 where appropriate.
- A justified architecture pattern and preliminary API/data contract.
- A feasible semester scope, resource estimate, risk plan, and measurable client outcome.
- No production deployment, sensitive-data collection, or expense commitment without written approval.

SHARED TEAM AND CLIENT PROFILE

Field	Required information
Client organization	Brialyns Art Sign, P. Guevara Ave, Santa Cruz, Laguna Kaisarap, Laguna University, Brgy. Bubukal, Santa Cruz, Laguna Carbonations, Laguna University, Brgy. Bubukal, Santa Cruz, Laguna
Industry/sector	Printing Services / Advertising & Custom Print Shop Food & Beverage / Food Stall (Selling Wapizza and Manju Drinks/Beverage / Student-Run Drink Stall
Authorized client representative	Jena Bersamina, Owner Business Owner, Benedict Monsanto, Mikayla Arban, Leader Lance Espital (Co-owner / Student Business Representative)
Primary process for study	Stock material tracking, customer order management, and order scheduling Daily ordering, cashiering, shift cash accounting, sales recording, and customer loyalty management Manual order-taking, per-cup sales counting, cash handling, and ingredient inventory tracking
Initial consultation date	August, 06 2026, Online interview conducted by Jane Angelie M. Ventura and Lance Bradly A. Monahan August 09, 2026 Online interview conducted by Jane Angelie M. Ventura and Lance Bradly A. Monahan August 07, 2026 Online interview conducted by Jane Angelie M. Ventura and Lance Bradly A. Monahan



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Field	Required information
Evidence of willingness	Completed Activity 1 interview documentation and questionnaire Completed Activity 1 interview documentation and questionnaire Completed Activity 1 interview documentation and questionnaire
Client constraints	[Schedule, access, data, branding, equipment, or policy limitations] Manual operations, reliance on paper logbooks/direct messaging, no automated physical printing, no online payment gateway integration, and no delivery service management No changes to food preparation/cooking processes; no hardware (receipt printers/order displays); manual entry for e-wallet reference numbers; no delivery service or online ordering. Manual store operations rely on paper logbooks, no hardware/IoT sensor integration, no online/e-commerce customer ordering, no third-party digital payment gateway API integration and no delivery or logistics tracking.



TITLE PROPOSAL 1

PROPOSED TITLE: An Integrated Mobile Point-of-Sale, IoT-Based Inventory Monitoring, and Order-Workflow System for Custom Printing Businesses

I. PROJECT OVERVIEW AND RATIONALE

[Introduce the client, existing workflow, verified problem, affected users, and why an integrated solution is needed. Support major claims with interview evidence or credible references.]

Brialyns Art Sign is a custom print shop owned by Jena Bersamina. The business provides tarpaulins, invitations, giveaways, screen prints, and signs. Currently, the owner and staff process customer orders, handle pricing discussions, and track inventory manually using paper logbooks, direct social media messages, and occasional Excel sheets.

Because there is no centralized tracking system, staff face difficulties knowing whether sufficient materials are in stock for incoming jobs and determining which orders to process first when handling different target completion dates. This manual process leads to excessive time spent answering customer inquiries, confused inventory management, and delayed order completions. An integrated solution combining a role-based mobile app for staff and a web dashboard for the owner is needed to digitize order creation, automate priority queues, track material stock levels, and provide visibility over shop operations.

To further address material shortages before they cause delays, the system will integrate an ESP32-based RFID check-out station for key printing materials to capture real-time usage events. Each material variant (e.g., a specific tarpaulin size, ink color, or paper type) is assigned one RFID tag, which staff tap whenever a unit of that material is taken from storage. This usage data feeds a predictive analytics layer that calculates dynamic reorder points and forecasts material demand, while a separate model estimates realistic order completion dates based on the shop's live workload. This moves the business from reactive stock-checking to proactive planning.

II. STATEMENT OF THE PROBLEM

[State the main operational problem as a clear question or problem statement. Identify two to four specific process or information gaps.]

Inefficient Order and Schedule Tracking- Reliance on paper logbooks and handwritten notes makes it difficult to track order status and completion target dates.

Lack of Order Prioritization- Staff have no automated system to identify which orders are overdue, urgent, or upcoming, causing confusion in production.

Manual Inventory Monitoring- Printing supplies (blank shirts, tarpaulin, mugs, paper) are not tracked automatically, leading to sudden material shortages during production.

No Real-Time Physical Stock Visibility- Even with digital order tracking, material counts are only as accurate as the last manual check, so the owner still discovers shortages late instead of getting advance warning based on actual usage.

No Realistic Completion Estimates- The shop has no way to predict how long an order will actually take based on current backlog and job complexity, leading to missed target dates and inconsistent customer expectations.



III. PROJECT OBJECTIVES

General objective

[One outcome-focused objective for the complete integrated system.]

To design and develop a client order tracking, production scheduling, inventory management, and predictive analytics system that helps a small printing business organize customer orders, prioritize production based on target dates, monitor the availability of printing materials in real time, and forecast material and completion needs.

Specific objectives

1. To develop a role-based mobile application that enables POS/Cashier staff to create orders, upload design layouts, record payments, and set target pick-up dates.
2. To provide a Production Staff mobile interface with a production queue that automatically prioritizes orders according to target dates.
3. To implement real-time order status tracking from Pending to In Production, Ready for Pickup, and Completed.
4. To track printing material inventory and display low-stock or insufficient-stock indicators for upcoming jobs.
5. To build a web-based Admin/Owner system featuring dashboards, user management, production schedules, and inventory monitoring.
6. To provide the Admin/Owner with production and inventory analytics, including order volume trends, on-time versus overdue completion rates, and material usage trends, along with a basic material restock forecast based on recent consumption trends.
7. To integrate an ESP32-based RFID check-out station, with one RFID tag assigned per material variant (e.g., tarpaulin size, ink color, paper type), to log real-time material usage events.
8. To implement a predictive inventory forecasting model that calculates dynamic Reorder Points (ROP) and forecasts material demand using time-series analysis (e.g., Holt-Winters or Exponential Smoothing) based on real-time RFID check-out data.
9. To develop a dynamic ETA feature that predicts realistic order completion dates based on live shop backlog, job complexity, and current operational capacity.



IV. CLIENT VALUE AND SUCCESS INDICATORS

Expected client outcome	Current baseline/evidence	Target or acceptance indicator
Orders are organized and easier to track	Based on Activity 1 client interview, orders are currently managed manually/informally, making it difficult to determine which orders should be prioritized (per Identified Problem)	All customer orders are recorded in the system with complete details (customer info, items, layout, target date, payment) and are retrievable through the Production Queue and Admin dashboard
Orders are processed according to urgency, reducing missed or late completions	No existing system flags overdue or urgent orders; prioritization is currently manual and inconsistent	The system automatically sorts and displays orders as Overdue, Urgent, or Upcoming based on target date, and Production Staff can identify which orders to process first without manual computation
Printing materials are monitored to avoid production delays	Order progress (pending, in production, ready, completed) is currently tracked informally, with no shared visibility between staff and owner	Every order shows a clear, updated status (Pending, In Production, Ready for Pickup, Completed) accessible in real time by Production Staff and Admin/Owner
Business owner has full visibility of operations for better decision-making	The Admin/Owner currently has no consolidated dashboard for orders, inventory, production, and reports	The Admin/Owner can view a real-time dashboard summarizing orders, production schedules, inventory levels, and generated reports at any time
Staff can use the system with minimal training	No formal system currently exists; staff rely on manual or informal processes	Usability requirement is met - staff with minimal technical background can perform core tasks (create order, update status, check stock) with little to no training, as targeted in the Non-Functional Requirements
The owner can make more informed production and restocking decisions	No existing analytics or forecasting tool is available; decisions on staffing and material restocking are currently based on guesswork	The dashboard provides descriptive analytics (order volume trends, on-time vs. overdue completion rates, material usage trends) and a basic material restock forecast, supporting more informed production and inventory decisions
Materials are reordered before they run out, not after	Stock is only checked manually; shortages are discovered mid-production	The RFID check-out station logs material usage as it happens; the system auto-flags a reorder point calculated from actual checkout trends
Customers get realistic, data-based completion dates	Target dates are currently estimated informally by staff	The system computes an ETA per order based on backlog, job complexity, and shop capacity, updated as conditions change



TITLE PROPOSAL 1 - SCOPE AND USERS

V. SCOPE, LIMITATIONS, AND EXCLUSIONS

In scope	Out of scope / limitation
Role-based mobile app for POS/Cashier (order creation, layout upload, payment recording, target date setting)	No online payment gateway integration
Mobile app interface for Production Staff (production queue viewing, priority sorting, layout checking, status updates)	No separate customer-facing mobile ordering app
Web-based Admin/Owner dashboard for system management, user management, inventory monitoring, and scheduling	No automated physical printing or machine operations
Order status updates (Pending, In Production, Ready for Pickup, Completed)	No delivery service or courier management
Inventory tracking for printing supplies with low/insufficient stock alerts	Physical production and customer handoff remain manual staff tasks
Production and inventory analytics (order volume trends, completion rate tracking, material usage trends, and basic material restock forecasting)	
ESP32-based RFID check-out station for material variants (e.g., tarpaulin sizes, ink colors, paper types)	No automatic purchasing or supplier ordering; the system only alerts, and reordering remains a manual decision
Predictive Reorder Point (ROP) forecasting using time-series analysis on RFID check-out history	Tracking is limited to whole-unit check-outs per material variant; the system does not measure partial quantities within a single unit
Dynamic ETA computation based on backlog, job complexity, and capacity	Forecasting models are estimates based on available data and are not guaranteed exact predictions

VI. PROJECT LOCALE AND STAKEHOLDERS

Stakeholder / user	Role in current process	Expected use of proposed system
Business Owner / Admin	Manages overall shop operations, discusses designs, calculates prices, and manually tracks stock and orders.	Uses the web dashboard to monitor orders, production schedules, inventory levels, user accounts, and analytics reports.
POS / Cashier Staff	Receives customer orders, takes details, records payments manually, and discusses target dates.	Uses the mobile app to create orders, upload design layouts, record payments, and set target completion dates.
Production Staff	Handles design execution, screen printing, tarpaulin printing, and order preparation manually.	Uses the mobile app to view the prioritized production queue, inspect design layouts, check target dates, and update order statuses.



TITLE PROPOSAL 1 - INTEGRATION DESIGN

VII. PROPOSED INTEGRATED WORKFLOW

[Write or draw the end-to-end journey. Identify who or what creates an event, what data moves, which component processes it, and what verified outcome follows.]

The workflow begins when a customer provides their order details, contact information, design requirements, and requested completion or pick-up date. The POS/Cashier, using the mobile application, creates the event by encoding this information into the system. This includes the customer's details, the items and quantity, the uploaded design layout, the recorded payment, and the target completion or pick-up date.

Once the order is saved, the system checks the required printing materials against current inventory. If sufficient stock is available, the order is confirmed as successfully created. If not, the system displays an insufficient stock indicator, prompting the Admin/Owner to restock. This is the first verified outcome of the flow, and it directly supports Functional Requirement 6, which covers monitoring inventory and flagging low or insufficient stock.

The confirmed order data is stored in the Centralized Database, which the Production Staff accesses through the mobile app's Production Queue. Per Functional Requirement 3, the system automatically prioritizes orders based on their target dates, identifying which are overdue, urgent, or upcoming, so Production Staff can determine which orders to process first. As work progresses, the Production Staff updates the order status through the defined flow of Pending, then In Production, then Ready for Pickup, then Completed, as described in Functional Requirement 5. Each status update is saved and synced back to the database, serving as the verified outcome that confirms production progress.

In parallel, the Admin/Owner, through the web-based system, continuously monitors the same Centralized Database, viewing orders, production schedules, inventory, users, and reports through the dashboard, as described in Functional Requirement 7. This allows the Admin/Owner to oversee overall operations and respond to alerts such as low-stock indicators.

In parallel with the order and inventory workflow, staff tap the RFID tag of a material variant (e.g., Tarpaulin - Medium, Ink - Black) at the ESP32-based check-out station whenever a unit of that material is taken from storage. Each tap is logged as a check-out event and transmitted to the backend in real time, which deducts one unit from that variant's tracked stock count. The backend feeds this accumulated check-out history into a time-series forecasting model that calculates a dynamic reorder point per material variant, factoring in recent consumption trends. When stock approaches this calculated threshold, the system generates a reorder alert on the Admin/Owner dashboard, ahead of an actual stockout rather than after it.

Separately, each time an order is created or its status changes, the backend recalculates that order's estimated completion date (ETA) using the current production backlog, the complexity of the job, and available staff capacity. This ETA is shown to Production Staff and the Admin/Owner, and updates automatically as the queue shifts.

Once an order reaches Completed status, the responsibility shifts back to the business staff to physically hand over the finished item to the customer, since actual printing, production, and customer pick-up remain manual staff responsibilities, as stated in the system's Limitations.



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Throughout this journey, the Centralized Database functions as the single source of truth. Every update made by the POS/Cashier, Production Staff, Admin/Owner, or the RFID check-out station is reflected in real time across the mobile and web platforms, supported by regular backups and role-based access restrictions, as described in the Security and Reliability non-functional requirements.



VIII. PLATFORM RESPONSIBILITY MATRIX

Component	Primary responsibility	Key inputs	Key outputs
Web application	Allows the Admin/Owner to monitor and manage overall business operations - orders, production schedules, inventory, users, and reports through a dashboard	Data retrieved from the Centralized Database (orders, inventory levels, production status, user accounts, forecast data)	Dashboard views, generated reports (orders, inventory, production), production and inventory analytics (order volume trends, completion rates, material usage trends), basic material restock forecasts, reorder point alerts, order ETA views, system settings updates, user account management actions
Mobile application	Supports two roles: POS/Cashier (order creation, layout upload, payment recording) and Production Staff (production queue viewing, status updates)	Customer details, order items and quantity, design layout files, payment records, target completion/pick-up date, production status updates	New order records, updated order status (Pending, In Production, Ready for Pickup, Completed), stock-check requests, insufficient-stock alerts, order ETA display
Backend / API	Processes and validates data submitted by the mobile and web applications, applies business logic such as order prioritization, inventory checking, ROP forecasting, and ETA computation, and coordinates data flow to and from the database	Requests from mobile app (new orders, status updates), web app (monitoring/report requests), and ESP32 (RFID check-out events)	Processed and validated order records, prioritized production queue (Overdue, Urgent, Upcoming), inventory check results, calculated reorder points, forecasted material demand, computed order ETAs, synced data sent to both platforms
Shared database	Serves as the centralized, single source of truth for all order, customer, inventory, payment, user, RFID check-out, and log data	Data submitted from the mobile and web applications via the backend, and RFID check-out events from the ESP32 device	Consistent, real-time data made available to both the mobile and web applications; backup copies for data reliability
ESP32 / device	Reads RFID tags at the material check-out station and transmits each check-out event to the backend in real time	Unique material-variant ID read from the tapped RFID tag via the RC522 RFID reader module	Check-out events (material variant + timestamp) sent to Backend/API for stock deduction and demand forecasting
External service, if any	N/A	N/A	N/A



TITLE PROPOSAL 1 - DATA, DEVICE, AND FEASIBILITY

IX. PRELIMINARY API AND DATA CONTRACT

Flow / endpoint	Method	Example payload / shared record	Consumer
/api/auth/login	POST	{ "username": "cashier01", "password": "*****", "role": "POS_Cashier" }	Mobile app (POS/Cashier, Production Staff), Web app (Admin/Owner)
/api/orders (create order)	POST	{ "customer_name": "Juan Dela Cruz", "item_type": "T-shirt Printing", "quantity": 20, "layout_file": "layout01.png", "target_date": "2026-08-20", "payment_amount": 1500 }	POS/Cashier mobile app -> Backend/API -> Shared Database
/api/inventory/check	GET	{ "item_type": "T-shirt Printing", "required_qty": 20 } -> { "available": true, "current_stock": 45 }	Backend/API (triggered during order creation) -> POS/Cashier mobile app
/api/orders/queue	GET	{ "orders": [{ "order_id": "ORD-1023", "target_date": "2026-08-14", "priority": "Overdue" }] }	Production Staff mobile app
/api/orders/{order_id}/status	PUT/PATCH	{ "order_id": "ORD-1023", "status": "In Production" }	Production Staff mobile app -> Backend/API -> Shared Database
/api/inventory/alert	POST (system)	{ "item_type": "Tarpaulin", "current_stock": 2, "threshold": 5, "status": "Insufficient Stock" }	Backend/API -> Admin/Owner web app
/api/dashboard/summary	GET	{ "total_orders": 58, "pending": 12, "completed": 40, "low_stock_items": 3 }	Admin/Owner web app
/api/rfid/checkout	POST (device)	{ "material_variant_id": "TARP-MED", "tag_uid": "04A3B2C1", "sensor_id": "ESP32-01", "timestamp": "2026-08-20T09:00:00Z" }	ESP32 -> Backend/API -> Shared Database
/api/inventory/forecast	GET	{ "material_variant_id": "TARP-MED", "reorder_point": 4, "forecasted_demand_next_7_days": 9, "model": "Holt-Winters" }	Admin/Owner web app
/api/orders/{order_id}/eta	GET	{ "order_id": "ORD-1023", "estimated_completion": "2026-08-22", "based_on": ["backlog", "job_complexity", "capacity"] }	Production Staff mobile app, Admin/Owner web app
/api/reports/generate	GET	{ "report_type": "Inventory", "date_range": "2026-08-01 to 2026-08-14" } -> returns report file/data	Admin/Owner web app



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Flow / endpoint	Method	Example payload / shared record	Consumer
/api/sync (real-time data sync)	WebSocket/Polling	{ "event": "order_status_updated", "order_id": "ORD-1023", "new_status": "Ready for Pickup" }	Mobile app <-> Web app



IX-A. RECOMMENDED TECH STACK

Layer	Technology
Mobile App	Flutter (Dart)
Web Dashboard	Flutter Web
Backend/API	Node.js + Express (or Firebase Cloud Functions)
Database	Firebase Firestore
ESP32	Arduino/C++, RC522 RFID reader (13.56MHz, NTAG213 tags), WiFi + HTTP/Firebase client
Auth	Firebase Authentication (role-based: Cashier, Production Staff, Admin)

Why this combination: Flutter (mobile + web) with Firebase (auth, database, and real-time sync) is one of the more beginner-friendly yet legitimate stacks for capstone projects. It involves fewer moving parts, has extensive free tutorials, and directly supports the real-time and role-based features already outlined in this proposal.



X. ESP32 ROLE AND HARDWARE PLAN

Decision: ☒ Required because it senses/controls a physical process ☐ Optional enhancement ☐ Not applicable and not forced into scope

Sensor/input	Actuator/output	Sampling or trigger rule	Offline/error behavior
RC522 RFID reader module reading 13.56MHz NTAG213 tags; one tag is pre-assigned per material variant (e.g., tarpaulin size, ink color, paper type)	None (read-only station; no physical actuator in this release)	Reads and transmits a check-out event immediately when a tag is tapped; a short debounce delay prevents duplicate logging from a single tap	If WiFi is unavailable, check-out events are cached locally on the ESP32 and synced once connection is restored; the dashboard flags data as "stale" if no sync is received within the expected interval

XI. CLIENT CONFIRMATION FOR THIS ALTERNATIVE

The client representative confirms that the problem and proposed scope above were discussed. This is preliminary validation and does not authorize production deployment or financial commitment.

Client representative / signature	Date	Comments or required revisions

XII. FLOWCHART

[Lucidchart document]



TITLE PROPOSAL 2

PROPOSED TITLE: Kaisarap: A Mobile-Based Point-of-Sale and Sales Monitoring System with Digital Customer Loyalty Program

I. PROJECT OVERVIEW AND RATIONALE

[Introduce the client, existing workflow, verified problem, affected users, and why an integrated solution is needed. Support major claims with interview evidence or credible references.]

Kaisarap is a small food stall located at Laguna University that sells Wapizza and Manju. The daily operations are managed by the owner and cashier staff, who handle food preparation, order taking, payment collection, and customer loyalty activities. Currently, staff record customer orders manually in a handwritten paper logbook and tally daily sales by hand. Because there is no shift-based cash tracking, the owner regularly experiences unexplained cash shortages at the end of the day. Additionally, the stall relies on physical paper punch cards for its customer loyalty promo. These paper cards are easily lost or damaged, represent an unnecessary recurring printing expense, and cannot be easily tracked or verified. An integrated mobile POS and web system is needed to digitize sales recording, verify cash on hand per shift, and manage a product-specific digital loyalty program without altering how the food is prepared.

II. STATEMENT OF THE PROBLEM

[State the main operational problem as a clear question or problem statement. Identify two to four specific process or information gaps.]

Manual Sales Recording: Reliance on paper logbooks for recording orders and manually tallying sales leads to incorrect counts and unrecorded transactions.

Untraceable Cash Shortages: Lack of shift-based cash reconciliation makes it impossible for the owner to pinpoint when or why cash shortages occur during staff handovers.

Lack of Loyalty Program Visibility: Physical paper punch cards cannot be tracked or verified in real time.

III. PROJECT OBJECTIVES

General objective

[One outcome-focused objective for the complete integrated system.]

To design and develop a web and mobile sales monitoring and point-of-sale system with a shift-based cash accounting feature, a digital, product-specific loyalty rewards program, and sales analytics with basic demand forecasting for the client.

Specific objectives

1. To develop a mobile point-of-sale application that allows the cashier to record customer orders and process payments through cash, GCash, or PayMaya.
2. To implement a shift-based cash accounting feature that compares declared cash on hand against recorded cash sales and flags any shortage or overage.



3. To develop a digital loyalty feature that tracks stamps separately per product (Wapizza and Manju) and automatically identifies when a customer is eligible for a free item.
4. To automatically record and store all sales, shift, and loyalty data in a centralized database in real time.
5. To provide the business owner with a web-based dashboard for monitoring daily sales, cash variance per shift, and loyalty program activity.

IV. CLIENT VALUE AND SUCCESS INDICATORS

Expected client outcome	Current baseline/evidence	Target or acceptance indicator
Sales are recorded accurately and consistently, eliminating manual tallying errors	Client currently records sales manually, with no digital system in place to track daily transactions (per Identified Problem)	Every confirmed order is automatically saved to the database in real time, with no sale going unrecorded replacing the stall's handwritten sales logs
Cash shortages can be identified and traced per shift	The owner regularly experiences cash shortages at the end of the day but has no way of identifying when or how the shortage occurred, since there is no per-shift accounting	At the end of each shift, the system automatically compares declared cash on hand against recorded cash sales and immediately flags the shift as Short, Balanced, or Over, and by how much
The loyalty promo becomes trackable, verifiable, and less prone to loss or fraud	The stall currently uses physical punch cards per product, which are easily lost or damaged also for less cost, cannot be tracked or verified by staff, and give the owner no visibility into promo performance	Loyalty stamps are recorded digitally and automatically per confirmed purchase, tracked separately for Wapizza and Manju, with reward eligibility (10 stamps = 1 free item) automatically flagged and verifiable through a code at the counter
The owner gains real-time visibility into sales, cash, and loyalty performance	The owner currently has no consolidated way to monitor daily sales, cash discrepancies, or loyalty activity everything is manual and reactive	The owner can view a real-time web dashboard showing daily/monthly sales, per-shift cash variance, and loyalty program activity (total members, stamps issued, redemptions) at any time
The owner can make more informed sales and stocking decisions	No existing analytics or forecasting tool is available to the client; decisions are currently based on intuition rather than data	The dashboard provides descriptive analytics (best-selling products, peak hours, redemption trends) and a basic sales forecast per product based on recent sales trends, supporting more informed demand and stocking decisions
The digitized system does not disrupt the stall's existing food preparation process	The owner explicitly wants the cashiering and loyalty promo digitized without any change to how Wapizza or Manju are prepared or cooked (per Limitations)	The system covers only order encoding, payment, cash accounting, and loyalty tracking no feature alters or interferes with food preparation



TITLE PROPOSAL 2 - SCOPE AND USERS

V. SCOPE, LIMITATIONS, AND EXCLUSIONS

In scope	Out of scope / limitation
Mobile app for Cashiers (order encoding, payment processing, shift cash declaration)	No changes to the food preparation or cooking processes for Wapizza or Manju
Mobile app interface for Customers (loyalty registration, stamp tracking per product, reward redemption codes)	No automated verification of GCash/PayMaya transactions (reference numbers are entered manually)
Web-based Admin/Owner dashboard (sales monitoring, cash variance review, user management, descriptive analytics, and basic forecasting)	No online ordering or food delivery service management
Shift-based cash reconciliation (computing expected vs. declared closing cash to flag Short/Balanced/Over)	No multi-branch or multi-location management
Dual-product digital loyalty tracking	No physical hardware integration (such as receipt printers, barcode scanners, or order display monitors)
	No prescriptive analytics, such as automated restocking or promo recommendations.

VI. PROJECT LOCALE AND STAKEHOLDERS

Stakeholder / user	Role in current process	Expected use of proposed system
Business Owner (Client)	manages daily sales and handles manual business accounting.	Uses the web dashboard to monitor daily/monthly sales, review per-shift cash discrepancies, view analytics/forecasts, and manage staff accounts.
Cashier / Staff	Takes customer orders, writes details in a paper logbook, receives cash payments, and stamps physical loyalty cards.	Uses the mobile app to encode orders, calculate change, enter e-wallet reference numbers, declare shift cash, and verify customer redemption codes.
Customer	Places orders for Wapizza or Manju, pays for items, and collects physical paper loyalty stamps.	



TITLE PROPOSAL 2 - INTEGRATION DESIGN

VII. PROPOSED INTEGRATED WORKFLOW

[Write or draw the end-to-end journey. Identify who or what creates an event, what data moves, which component processes it, and what verified outcome follows.]

The workflow begins when a customer arrives at the stall to order Wapizza or Manju. The Cashier, after logging in to the mobile application using an authorized account, creates the event by selecting the ordered items and confirming the order. The data involved at this point includes the item selected, the quantity, and the total amount due.

Once the order is confirmed, the Cashier accepts payment through cash, GCash, or PayMaya. If the payment is cash, the system automatically computes and displays the change. If the payment is GCash or PayMaya, the Cashier manually enters the reference number provided by the customer, creating a traceable payment record, as required by Functional Requirement 3. This step produces the first verified outcome of the flow, which is a confirmed and recorded sale.

If the customer is a registered loyalty member, the system automatically adds a stamp to their Wapizza or Manju count, with each product tracked separately and never combined, as described in Functional Requirement 6. The system then checks the updated stamp count. If the count is less than ten, the customer continues earning stamps toward the next reward. If the count reaches ten, the customer becomes eligible for one free item of their choice, and the system notifies the customer of this eligibility, following Functional Requirement 7. The customer may later redeem this reward by presenting a code at the counter, which the Cashier verifies before releasing the free item.

At the start and end of every shift, the Cashier declares the cash on hand and the cash counted. The system automatically computes the expected cash based on recorded cash sales for that shift and compares it against the declared closing cash, as specified in Functional Requirement 5. This comparison produces a verified outcome showing whether the shift is Short, Balanced, or Over, and by how much, allowing the owner to trace the exact shift where any discrepancy occurred.

All order, payment, loyalty, and shift data generated by the Cashier and Customer interactions are automatically saved to the Centralized Database in real time, following Functional Requirement 8, removing the need for manual, handwritten sales and cash recording.

In parallel, the Business Owner, through the web based dashboard, continuously accesses the same Centralized Database to monitor daily and monthly sales, review cash variance per shift, and track loyalty program activity such as total members, stamps issued, and redemptions. The dashboard also presents descriptive analytics, including best selling products, peak hours, and loyalty redemption trends, along with a basic sales forecast per product based on recent sales trends, supporting more informed stocking and sales decisions.

Throughout this journey, the Centralized Database functions as the single source of truth. Every sale, payment, shift record, and loyalty update is stored and made accessible in real time to both the mobile application and the web dashboard, supported by regular backups and role based access control that separates cashier functions from owner functions, consistent with the system's Security and Reliability requirements. Notably, none of these processes alter or interfere with how Wapizza or Manju are actually prepared, in line with the stated Limitations of the system.



VIII. PLATFORM RESPONSIBILITY MATRIX

Component	Primary responsibility	Key inputs	Key outputs
Web application	Allows the Business Owner to monitor daily/monthly sales, review cash variance per shift, track loyalty program activity, view analytics and forecasts, manage cashier accounts, and configure system settings through a dashboard	Data retrieved from the Shared Database (sales records, shift cash comparisons, loyalty stamp data, user accounts)	Dashboard views, sales/cash/loyalty reports, exported reports (PDF/Excel), demand forecast summaries, cashier account management actions, system/product/price settings updates
Mobile application	Supports two roles, Cashier (order encoding, payment processing, shift cash declaration) and Customer (loyalty registration, stamp tracking, reward redemption)	Order details (items, quantity), payment method and amount, GCash/PayMaya reference numbers, declared opening/closing cash, customer login/registration data, redemption code requests	Confirmed order and sale records, computed change (for cash), shift cash declarations, loyalty stamp updates, reward eligibility notifications, redemption codes shown at counter
Backend / API	Processes and validates data submitted by the mobile and web applications, applies business logic such as change computation, stamp counting, shift cash comparison, and reward eligibility checks, and coordinates data flow to and from the database	Requests from the mobile app (new orders, payment confirmations, shift declarations, stamp updates) and web app (monitoring/report/forecast requests)	Validated sale and shift records, computed cash variance (Short/Balanced/Over), updated loyalty stamp counts, reward eligibility flags, generated analytics and forecast data, synced data sent to both platforms
Shared database	Serves as the centralized, single source of truth for all order, sales, payment, shift, loyalty, user, and product data	Data submitted from both the mobile and web applications via the backend	Consistent, real-time data made available to both the mobile and web applications; backup copies for data reliability and recovery
ESP32 / device	N/A	N/A	N/A



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Component	Primary responsibility	Key inputs	Key outputs
External service, if any	N/A	N/A	N/A



TITLE PROPOSAL 2 - DATA, DEVICE, AND FEASIBILITY

IX. PRELIMINARY API AND DATA CONTRACT

Flow / endpoint	Method or protocol	Example payload / shared record	Consumer
/api/auth/login	POST	{ "username": "cashier01", "password": "*****", "role": "Cashier" }	Mobile app (Cashier, Customer), Web app (Owner)
/api/customer/register	POST	{ "customer_name": "Ana Reyes", "contact": "0917xxxxxx", "password": "*****" }	Customer mobile app → Backend/API → Shared Database
/api/shift/start	POST	{ "cashier_id": "C-01", "shift_start": "2026-08-14T08:00:00Z", "declared_cash_on_hand": 2000 }	Cashier mobile app → Backend/API → Shared Database
/api/orders (create order)	POST	{ "cashier_id": "C-01", "items": [{ "product": "Wapizza", "qty": 2 }, { "product": "Manju", "qty": 1 }], "total_amount": 240, "payment_method": "Cash" }	Cashier mobile app → Backend/API → Shared Database
/api/orders/payment/cash	POST	{ "order_id": "ORD-2045", "amount_tendered": 300 } → response: { "change": 60 }	Backend/API → Cashier mobile app
/api/orders/payment/ewallet	POST	{ "order_id": "ORD-2046", "payment_method": "GCash", "reference_number": "GC123456789" }	Cashier mobile app → Backend/API → Shared Database
/api/loyalty/stamp	POST (system-triggered)	{ "customer_id": "CU-014", "product": "Wapizza", "order_id": "ORD-2045" } → response: { "current_stamp_count": 7 }	Backend/API → Customer mobile app
/api/loyalty/eligibility	GET	{ "customer_id": "CU-014", "product": "Manju" } → response: { "stamp_count": 10, "eligible": true, "redemption_code": "MJ-9F3K2" }	Customer mobile app
/api/loyalty/redeem	POST	{ "redemption_code": "MJ-9F3K2", "cashier_id": "C-01" } → response: { "status": "Redeemed", "product": "Manju" }	Cashier mobile app → Backend/API → Shared Database



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Flow / endpoint	Method or protocol	Example payload / shared record	Consumer
/api/shift/end	POST	{ "cashier_id": "C-01", "shift_end": "2026-08-14T17:00:00Z", "declared_cash_counted": 5300 }	Cashier mobile app → Backend/API → Shared Database
/api/shift/reconciliation	GET (system-computed)	{ "shift_id": "SH-114", "expected_cash": 5350, "declared_cash": 5300, "variance": -50, "status": "Short" }	Backend/API → Owner web app
/api/dashboard/summary	GET	{ "daily_sales": 8400, "total_orders": 62, "shortage_alerts": 1, "loyalty_redemptions_today": 2 }	Owner web app
/api/sales/history	GET	{ "transactions": [{ "order_id": "ORD-2045", "product": "Wapizza", "amount": 160, "payment_method": "Cash" }] }	Owner web app
/api/analytics/descriptive	GET	{ "best_selling": "Wapizza", "peak_hours": "11AM-1PM", "redemption_trend": "increasing" }	Owner web app
/api/analytics/forecast	GET	{ "product": "Manju", "forecast_next_period": 180, "based_on": "last_30_days_sales" }	Owner web app
/api/users (cashier account management)	GET / POST / PUT	{ "user_id": "C-02", "name": "Mark Villanueva", "role": "Cashier", "status": "Active" }	Owner web app → Backend/API → Shared Database
/api/settings	GET / PUT	{ "products": ["Wapizza", "Manju"], "loyalty_threshold": 10, "payment_methods": ["Cash", "GCash", "PayMaya"] }	Owner web app → Backend/API → Shared Database



Flow / endpoint	Method or protocol	Example payload / shared record	Consumer
/api/reports/export	GET	{ "report_type": "Sales", "format": "PDF", "date_range": "2026-08-01 to 2026-08-14" } → returns file	Owner web app
/api/notifications	POST (system-triggered)	{ "type": "Cash Shortage Alert", "shift_id": "SH-114", "amount": -50 }	Backend/API → Owner web app
/api/sync (real-time data sync)	WebSocket / Polling	{ "event": "stamp_added", "customer_id": "CU-014", "product": "Manju", "new_count": 10, "timestamp": "2026-08-14T10:32:00Z" }	Mobile app ↔ Web app (via Backend/API and Shared Database)

X. ESP32 ROLE AND HARDWARE PLAN

Decision: ☐ Required because it senses/controls a physical process ☐ Optional enhancement ☐ Not applicable and not forced into scope

Sensor/input	Actuator/output	Sampling or trigger rule	Offline/error behavior
N/A	N/A	N/A	N/A

XI. CLIENT CONFIRMATION FOR THIS ALTERNATIVE

The client representative confirms that the problem and proposed scope above were discussed. This is preliminary validation and does not authorize production deployment or financial commitment.

Client representative / signature	Date	Comments or required revisions

XII. FLOWCHART

 Lucidchart document



TITLE PROPOSAL 3

PROPOSED TITLE: Carbonations: An Integrated Mobile POS and Web-Based Sales-Inventory Monitoring System for Small Drink Stall Businesses

I. PROJECT OVERVIEW AND RATIONALE

[Introduce the client, existing workflow, verified problem, affected users, and why an integrated solution is needed. Support major claims with interview evidence or credible references.]

Carbonations is a small drink stall business operating at Laguna University, managed by student co-owner Lance Espital and nine team members who rotate daily duties. The stall serves various soda-based, milk-based, and water-based beverages in different flavors.

Currently, the assigned cashier manually records every order, counts sold cups, and tracks ingredient inventory using paper logbooks and handwritten notebooks. During peak campus hours, heavy customer crowds cause panic and confusion among staff. Manual recording results in slow checkout, long customer lines, payment calculation errors, and inaccurate stock tracking. Furthermore, the owner frequently runs out of key ingredients (like syrups, cups, and ice) without advance warning. An integrated mobile POS app and web dashboard is needed to speed up order processing, automate recipe-based stock deduction, and provide real-time sales reporting.

II. STATEMENT OF THE PROBLEM

[State the main operational problem as a clear question or problem statement. Identify two to four specific process or information gaps.]

Manual Order and Sales Recording: Relying on paper logbooks for recording orders and manually tallying cup sales causes slow transactions, customer delays, and incorrect daily sales calculations.

Untracked Ingredient Inventory: Lack of real-time inventory tracking causes ingredient shortages (syrups, powders, cups) to go unnoticed until items are completely out of stock during customer orders.

Inaccurate Business Reporting and Planning: The owner lacks centralized sales logs, peak-hour analytics, and demand projections to make informed ingredient stocking decisions.

III. PROJECT OBJECTIVES

General objective

[One outcome-focused objective for the complete integrated system.]

To design and develop a web-based sales monitoring and inventory management system with an integrated mobile point-of-sale application for Carbonations to automate transaction logging and ingredient stock management.

Specific objectives

1. To develop a mobile POS app that allows staff to quickly record customer drink orders and compute payment/change automatically.



2. To build an automatic inventory feature that deducts ingredient stock levels based on predefined drink recipes upon order confirmation.
3. To provide a web-based dashboard for the owner to monitor real-time daily sales, view current stock levels, and receive low-stock alerts.
4. To implement descriptive sales analytics (best-selling drinks, peak sales hours) and basic demand forecasting per product to assist stocking decisions.
5. To secure all transaction logs and inventory records in a centralized cloud database for searchable order history and report generation.

IV. CLIENT VALUE AND SUCCESS INDICATORS

Expected client outcome	Current baseline/evidence	Target or acceptance indicator
Faster order checkout and automated change calculation	Manual entry in paper logbooks; manual calculation of order totals and change.	100% automated calculation of order totals and change on mobile POS.
Real-time stock deduction and inventory tracking	Un-tracked ingredient stock leading to unexpected out-of-stock items.	Automatic deduction of ingredients per order confirmation and instant low-stock alerts.
Accurate sales history and reporting	Handwritten notebook logs that cause inaccurate cup counts and lost sales data.	100% automated sales report generation and searchable digital transaction logs.



TITLE PROPOSAL 3 - SCOPE AND USERS

V. SCOPE, LIMITATIONS, AND EXCLUSIONS

In scope	Out of scope / limitation
Mobile POS application for stall staff (drink item selection, order building, cash payment confirmation, receipt generation)	No online customer ordering, mobile delivery app integration, or e-commerce features
Web-based admin dashboard for the business owner (sales monitoring, inventory tracking, recipe management, staff account controls)	No third-party digital payment gateway API integration (e-wallet transactions verified manually)
Automatic recipe-based ingredient stock deduction upon order confirmation	No IoT or hardware sensor-based automatic physical stock detection
Low-stock notifications and printable sales/inventory report generation	No multi-branch or multi-stall location management functionality
Descriptive sales analytics (best-sellers, peak hours) and basic demand forecasting	No customer-facing mobile application or customer loyalty program features

VI. PROJECT LOCALE AND STAKEHOLDERS

Stakeholder / user	Role in current process	Expected use of proposed system
Business Owner / Admin (Lance Espital & Co-owners)	manages inventory, reviews paper sales logs, and handles business planning.	Uses the web dashboard to monitor live sales, track ingredient stock levels, receive low-stock alerts, review analytics/forecasts, and manage staff accounts.
Stall Staff / Cashier	Takes customer drink orders, writes sales in notebooks, calculates change by hand, and counts cup sales.	Uses the mobile POS app to encode drink orders, compute total prices and change automatically, and confirm orders to deduct stock.



TITLE PROPOSAL 3 - INTEGRATION DESIGN

VII. PROPOSED INTEGRATED WORKFLOW

[Write or draw the end-to-end journey. Identify who or what creates an event, what data moves, which component processes it, and what verified outcome follows.]

The workflow begins when a customer orders a drink at the stall. The Stall Staff/Cashier, using the mobile POS app, creates the event by selecting the drink items and quantities, then confirming the order. The data involved at this point includes the items selected, the quantity, and the total amount due.

Once the order is confirmed, the Cashier accepts payment through cash, GCash, or PayMaya. If the payment is cash, the system automatically computes and displays the change. If the payment is GCash or PayMaya, the Cashier manually enters the reference number provided by the customer, creating a traceable payment record, since the system does not verify e-wallet payments automatically. This step produces the first verified outcome of the flow, which is a confirmed and recorded sale.

Upon order confirmation, the system automatically deducts the ingredients used from inventory based on the predefined recipe of each drink item. The system then checks whether any ingredient's remaining stock has fallen below a set threshold. If it has, the system notifies the Admin/Owner with a low-stock alert, giving the owner advance warning before an ingredient runs out completely.

All order, sales, payment, and inventory data generated by the Cashier's actions are automatically saved to the Shared Database in real time, removing the need for manual, handwritten sales and stock tallying.

In parallel, the Business Owner/Admin, through the web-based dashboard, continuously accesses the same Shared Database to monitor daily, weekly, and monthly sales, view current ingredient stock levels, review order history, and respond to low-stock alerts. The dashboard also presents descriptive analytics, including best-selling drinks, peak sales hours, and ingredient usage trends, along with a basic demand forecast per product based on recent sales trends, supporting more informed stocking decisions.

Throughout this journey, the Shared Database functions as the single source of truth. Every order, payment, and inventory update made by the Stall Staff/Cashier is reflected in real time on the web dashboard, supported by regular backups and role-based access control that separates staff functions from admin functions. Notably, none of these processes involve any physical hardware or IoT sensors; ingredient deduction is entirely recipe-based rather than sensor-based, consistent with the system's stated Limitations.



VIII. PLATFORM RESPONSIBILITY MATRIX

Component	Primary responsibility	Key inputs	Key outputs
Web application	Allows the Business Owner/Admin to monitor sales, inventory, and analytics, and to manage staff accounts through a dashboard	Data retrieved from the Shared Database (orders, sales records, inventory levels, staff accounts)	Dashboard views, sales/inventory reports, low-stock notifications, descriptive and predictive analytics, staff account management actions
Mobile application	Supports the Stall Staff/Cashier role, order encoding, payment processing (cash, GCash, PayMaya), and order confirmation	Drink items and quantities, payment method and amount, GCash/PayMaya reference numbers	Confirmed order and sale records, computed change (for cash), stock-deduction triggers
Backend / API	Processes and validates data submitted by the mobile and web applications, applies business logic such as recipe-based stock deduction, low-stock checking, and analytics computation	Requests from the mobile app (new orders, payment confirmations) and web app (monitoring/report/analytics requests)	Validated order and sale records, updated inventory levels, low-stock alerts, generated analytics and forecast data, synced data sent to both platforms
Shared database	Serves as the centralized, single source of truth for all order, sales, payment, inventory, recipe, and user data	Data submitted from the mobile and web applications via the backend	Consistent, real-time data made available to both platforms; backup copies for data reliability
ESP32 / device	N/A	N/A	N/A
External service, if any	N/A	N/A	N/A



TITLE PROPOSAL 3 - DATA, DEVICE, AND FEASIBILITY

IX. PRELIMINARY API AND DATA CONTRACT

Flow / endpoint	Method or protocol	Example payload / shared record	Consumer
/api/auth/login	POST	{ "username": "staff01", "password": "*****", "role": "Staff" }	Mobile app (Staff), Web app (Admin)
/api/orders (create order)	POST	{ "staff_id": "S-01", "items": [{ "product": "Iced Tea", "qty": 2 }], "total_amount": 90, "payment_method": "Cash" }	Mobile app → Backend/API → Shared Database
/api/orders/payment/cash	POST	{ "order_id": "ORD-3012", "amount_tendered": 100 } → response: { "change": 10 }	Backend/API → Mobile app
/api/orders/payment/ewallet	POST	{ "order_id": "ORD-3013", "payment_method": "GCash", "reference_number": "GC998877" }	Mobile app → Backend/API → Shared Database
/api/inventory/deduct	POST (system-triggered)	{ "order_id": "ORD-3012", "ingredients_deducted": { "syrup": 30, "cup": 1, "ice": 1 } }	Backend/API → Shared Database
/api/inventory/alert	POST (system-triggered)	{ "ingredient": "syrup", "current_stock": 3, "threshold": 5, "status": "Low Stock" }	Backend/API → Web app (notification)
/api/dashboard/summary	GET	{ "daily_sales": 4200, "total_orders": 58, "low_stock_items": 2 }	Web app
/api/orders/history	GET	{ "orders": [{ "order_id": "ORD-3012", "product": "Iced Tea", "amount": 90 }] }	Web app
/api/analytics/descriptive	GET	{ "best_selling": "Iced Tea", "peak_hours": "10AM-12PM", "ingredient_usage": { "syrup": "high" } }	Web app
Web app	GET	{ "product": "Iced Tea", "forecast_next_period": 210, "based_on": "last_30_days_sales" }	Web app
/api/users (staff account management)	GET / POST / PUT	{ "user_id": "S-02", "name": "Rey Santos", "role": "Staff", "status": "Active" }	Web app → Backend/API → Shared Database
/api/reports/generate	GET	{ "report_type": "Sales", "date_range": "2026-08-01 to 2026-08-14" } → returns file/data	Web app
/api/sync (real-time data sync)	WebSocket / Polling	{ "event": "inventory_updated", "ingredient": "syrup", "new_stock": 47, "timestamp": "2026-08-14T10:32:00Z" }	Mobile app ↔ Web app (via Backend/API and Shared Database)



X. ESP32 ROLE AND HARDWARE PLAN

Decision: ☐ Required because it senses/controls a physical process ☐ Optional enhancement ☐ Not applicable and not forced into scope

Sensor/input	Actuator/output	Sampling or trigger rule	Offline/error behavior
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A

XI. CLIENT CONFIRMATION FOR THIS ALTERNATIVE

The client representative confirms that the problem and proposed scope above were discussed. This is preliminary validation and does not authorize production deployment or financial commitment.

Client representative / signature	Date	Comments or required revisions

XII. FLOWCHART

 Lucidchart document



TITLE COMPARISON AND RECOMMENDATION

Score each title using the same evidence. A feasible, integrated, client-valuable project should rank higher than a larger but weakly connected feature set.

Criterion	Weight	Title 1	Title 2	Title 3
Client need and evidence	20%			
Integration depth and coherence	25%			
Web/Mobile role clarity	15%			
ESP32 relevance and feasibility	15%			
Semester feasibility and resources	10%			
Security, privacy, and ethical fit	10%			
Client willingness and access	5%			
TOTAL	100%			

TEAM RECOMMENDATION

Recommended title: _____

Justification based on the comparison, client feedback, feasibility, and integration value:

PROPOSED TEAM RESPONSIBILITIES

Member	Primary ownership	Secondary/cross-training responsibility	Evidence of contribution

DECLARATION

We certify that the information in this proposal is based on our preliminary investigation and client consultation. We understand that title approval does not authorize collection of sensitive data, production deployment, expenses, or use of client resources beyond written permissions.

Student name and signature	Student name and signature



FACULTY EVALUATION AND APPROVAL

Decision	Mark
Approved title proposal	☐ Title 1 ☐ Title 2 ☐ Title 3
Approved with revisions	☐
For re-presentation	☐
Not approved	☐

Required revisions / conditions

APPROVAL SIGNATURES

Role	Printed name and signature	Date
SIA 2 Subject Teacher		
Mobile Application Development Teacher		
Client Representative		
Program Chair / Coordinator		
Dean, College of Computing Studies (Noted)		

FINAL APPROVED PROJECT TITLE

After approval, the team must freeze the minimum viable scope, revise the architecture and API contract, and establish a shared repository, issue tracker, integration test plan, and client checkpoint schedule before development begins.